

Lecture 7: Time Series
Tasks, Chaos and Signal Characteristics
Linear Models and NVAR

Deep Learning in Classical and Quantum Physics

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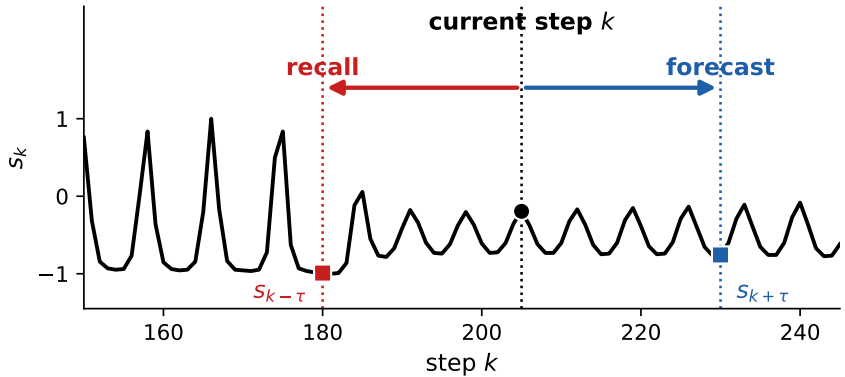
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Time-Series Tasks



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Time-Series Tasks: Recall or Forecast



At the current step k the model has seen $s_1, \ldots, s_k$, one number per step.

- **recall**: target $y_k = s_{k-\tau}$, a value τ steps in the past, which tests memory
- **forecast**: target $y_k = s_{k+\tau}$, a value τ steps ahead, τ is the horizon

A Measured Signal from a Dynamical System

Definition (Time series)

Measurements $s_1, \ldots, s_N$ at equal intervals Δt,

$$s_k = h(\mathbf{x}(k\Delta t)) + \eta_k$$

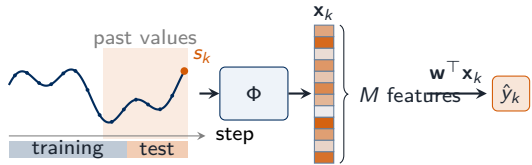
$\mathbf{x}(t) \in \mathbb{R}^d$: state of the system, not observed; h: measurement function; η_k: measurement noise; k: step

- we record one number per step, never the full state $\mathbf{x}$
- examples are a laser intensity, a convection mode or a blood-cell count
- before forecasting we **characterise** the signal, its type of dynamics, its memory and its predictability time

M. Płodzień, *Lecture Notes on Time Series Forecasting. A Dynamical-Systems Approach*, github.com/MarcinPlodzien/time-series-forecasting-book

Past Values, Features and a Linear Readout

$$(s_1, \ldots, s_k) \xrightarrow{\Phi} \mathbf{x}_k \in \mathbb{R}^M \xrightarrow{\text{linear}} \hat{y}_k = \mathbf{w}^\top \mathbf{x}_k$$



- Φ: M features, linear (the past values) or nonlinear (their products), of the last L values or a state $\mathbf{x}_k = F(\mathbf{x}_{k-1}, s_k)$
- in AR, NVAR and reservoirs Φ is fixed and only the readout $\mathbf{w} \in \mathbb{R}^M$ is trained, by least squares
- recurrent neural networks (Lecture 8) train Φ and $\mathbf{w}$ together

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The Forecasting Problem



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Forecast, Horizon and Window

Definition (Forecast at horizon τ)

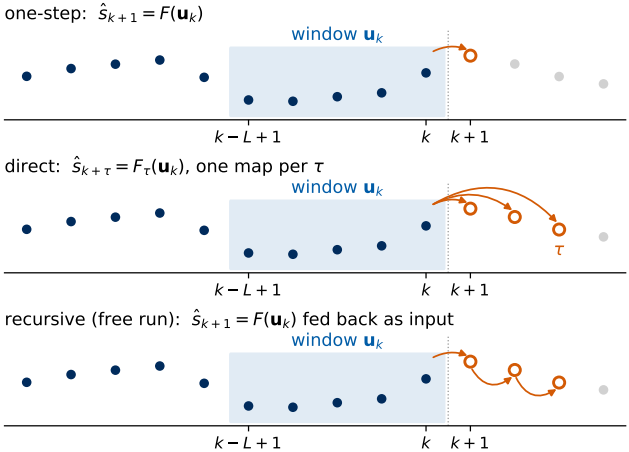
$$\hat{s}_{k+\tau} = F_\tau(\mathbf{u}_k), \qquad \mathbf{u}_k = (s_k, s_{k-1}, \ldots, s_{k-L+1})$$

τ: horizon in samples; $\mathbf{u}_k$: window of the last L values; F_τ: predictor fitted on past data

- the window stands in for the unobserved state $\mathbf{x}$
- we use horizons $\tau = 1, \ldots, 50$ throughout

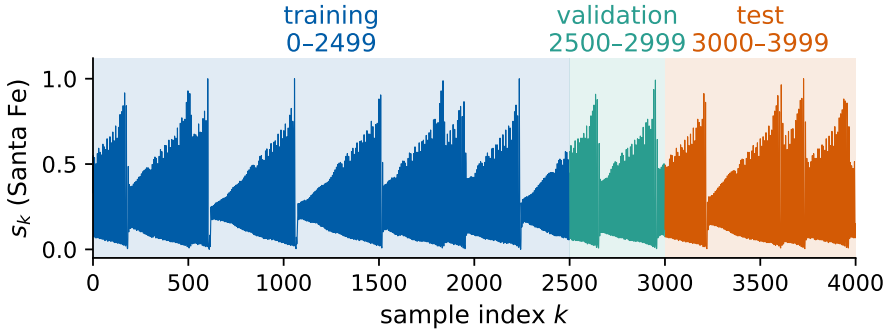
J. D. Farmer, J. J. Sidorowich, Phys. Rev. Lett. 59, 845 (1987), doi:10.1103/PhysRevLett.59.845

One-Step, Direct and Recursive Forecasts



- **one-step**: predict $\hat{s}_{k+1}$ from measured values
- **direct**: one map F_τ per horizon
- **recursive**: feed the one-step map its own outputs, so errors accumulate

Training and Testing in Time Order



Santa Fe signal, rescaled to $[0, 1]$; every signal has $N = 4000$ samples split in this way.

- fit on the training block, choose settings on the validation block, report on the test block
- never shuffle: random windows would put test values into the training inputs

Normalised Error

Predictions $\hat{y}_i = \hat{s}_{k_i+\tau}$, targets $y_i = s_{k_i+\tau}$, over n test steps k_i.

Definition (Normalised mean-square error (NMSE))

$$\mathrm{NMSE}(\tau) = \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

$\bar{y}$: mean of the test targets

- $\mathrm{NMSE} = 0$ for a perfect forecast
- $\mathrm{NMSE} = 1$ for the constant forecast $\hat{y}_i = \bar{y}$, so a useful model stays below 1

Capacity

Definition (Capacity)

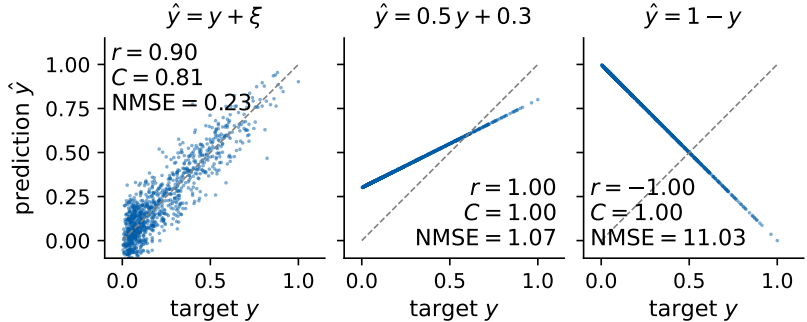
$$C(\tau) = r(\tau)^2, \qquad r(\tau) = \frac{\sum_i (\hat{y}_i - \bar{\hat{y}})(y_i - \bar{y})}{\sqrt{\sum_i (\hat{y}_i - \bar{\hat{y}})^2 \sum_i (y_i - \bar{y})^2}}, \qquad C_{\mathrm{tot}} = \sum_{\tau=1}^{50} C(\tau)$$

r: Pearson correlation of forecast and target; $C \in [0, 1]$

- $C(\tau)$ measures whether the forecast follows the shape of the target
- $C_{\mathrm{tot}} \in [0, 50]$ counts the effectively predictable horizons

J. Dambre *et al.*, Sci. Rep. 2, 514 (2012), doi:10.1038/srep00514

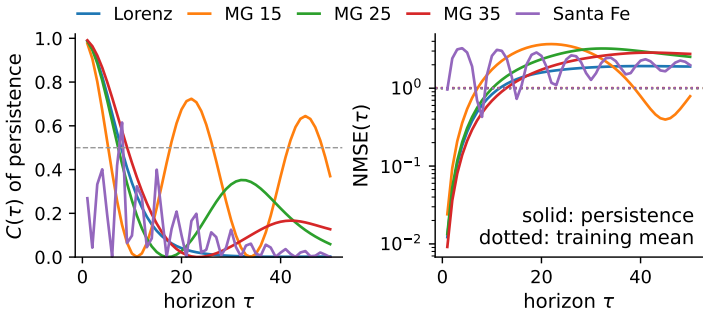
Capacity and Error Measure Different Things



Forecast against target on the Santa Fe test block, for three constructed forecasts.

- $\hat{y} = 0.5\,y + 0.3$: $C = 1.00$, $\mathrm{NMSE} = 1.07$, right shape, wrong amplitude
- so we always report C together with the NMSE

Persistence Baseline



Left $C(\tau)$ of persistence $\hat{s}_{k+\tau} = s_k$, right its NMSE (solid) and that of the training mean (dotted).

- persistence is strong for smooth signals at short τ, weak for Santa Fe
- every model must beat persistence and the mean

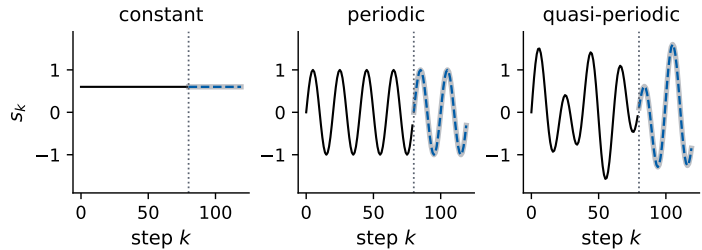
3

Easy, Chaotic and Random Signals



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Signals That Are Easy to Forecast



Past (black) up to $k = 80$, true continuation (grey) and forecast (dashed blue).

- **constant**: the last value, $\hat{s}_{k+\tau} = s_k$ (persistence)
- **periodic** with period P: the value one period earlier, $\hat{s}_{k+\tau} = s_{k+\tau-P}$
- **quasi-periodic**, two sines with an irrational frequency ratio: never repeats, yet obeys an exact linear recurrence $s_k = \sum_{j=1}^{4} a_j s_{k-j}$, fitted on the past and iterated

All three forecasts are exact to round-off, these are trivial tasks.

Deterministic Chaos

A known rule maps the state to the next state, without any randomness.

Definition (Dynamical system)

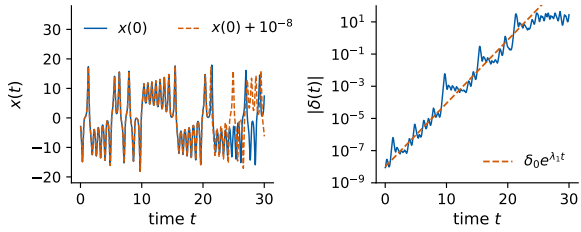
$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}) \quad \text{(flow)} \qquad \text{or} \qquad \mathbf{x}_{n+1} = \mathbf{f}(\mathbf{x}_n) \quad \text{(map)}$$

$\mathbf{x} \in \mathbb{R}^d$: point of the phase space; $\mathbf{F}$, $\mathbf{f}$: known equations

- in a **dissipative** system the motion settles on an **attractor**
- **chaos**: bounded, never-repeating motion in which nearby states separate exponentially
- the equations fix the future exactly, yet any initial error grows, so the far future cannot be forecast

S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed., Westview Press (2015)

Two Nearby States Separate



E. N. Lorenz, J. Atmos. Sci. 20, 130 (1963)

Lorenz-63 from two states 10^{-8} apart; left $x(t)$ of both, right their distance $|\delta(t)|$.

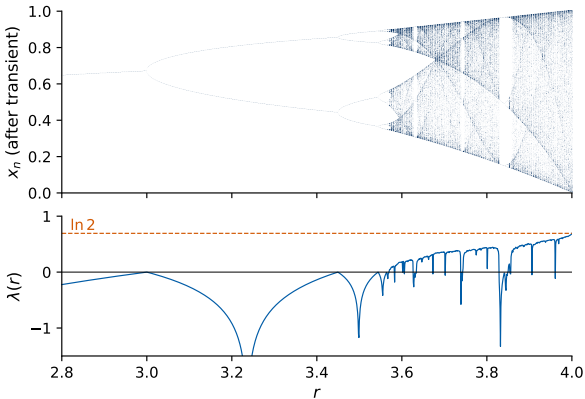
Definition (Largest Lyapunov exponent)

$$|\delta(t)| \approx |\delta_0|\, e^{\lambda_1 t}, \qquad \lambda_1 = \lim_{t\to\infty} \lim_{|\delta_0|\to 0} \frac{1}{t} \ln \frac{|\delta(t)|}{|\delta_0|}$$

$\delta(t)$: difference of the two states; $T_\lambda = 1/\lambda_1$: Lyapunov time

The pair agrees until $t \approx 20$; slope of $\ln|\delta|$ 0.93, average $\lambda_1 = 0.907$.

Lyapunov Exponent of the Logistic Map



$$x_{n+1} = r\,x_n(1 - x_n)$$

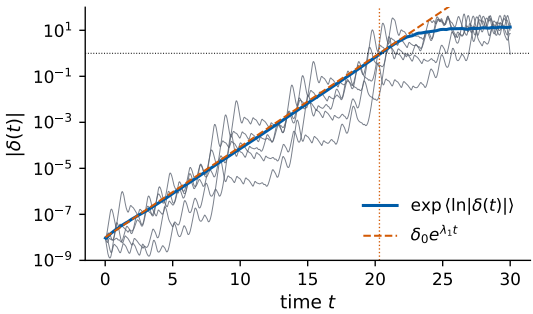
$$\lambda = \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \ln\left|f'(x_n)\right|$$

$f'(x) = r(1 - 2x)$; a small separation is multiplied by $|f'(x_n)|$ at each step

- upper panel the values visited, lower panel $\lambda(r)$
- $\lambda < 0$ for fixed points and cycles, $\lambda > 0$ for chaos, first at $r = 3.570$
- $\lambda(4) = 0.6931 = \ln 2$

R. M. May, Nature 261, 459 (1976), doi:10.1038/261459a0

The Prediction Horizon



$|\delta(t)|$ of five pairs (grey) and $\exp\langle \ln|\delta|\rangle$ over 200 pairs (blue).

An initial error δ_0 grows until it reaches a tolerance Δ,

$$T_{\text{pred}} = \frac{1}{\lambda_1} \ln \frac{\Delta}{\delta_0}$$

- Lorenz, $|\delta_0| = 10^{-8}$, $\Delta = 1$ is reached at $t = 20.4$, the formula gives 20.3
- a thousand times better data add only $\ln(1000)\, T_\lambda \approx 6.9\, T_\lambda$
- no model can beat this limit, only approach it

Lyapunov Spectrum

In d dimensions a small sphere of errors becomes an ellipsoid, and each axis grows at its own rate.

Definition (Lyapunov exponents)

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d, \qquad k\text{-dimensional volumes grow as } e^{(\lambda_1+\cdots+\lambda_k)t}$$

λ_1: length; $\lambda_1 + \lambda_2$: areas; $\lambda_1 + \cdots + \lambda_d$: volume of phase space

- $\lambda_1 > 0$ signals chaos, a flow has one zero exponent, and a dissipative system a negative sum
- Lorenz-63 has $\lambda = (0.907,\ 0.0001,\ -14.573)$ per time unit

The exponents are computed from the equations by following tangent vectors (Benettin algorithm, Lab 033). V. I. Oseledets, Trans. Moscow Math. Soc. 19, 197 (1968); G. Benettin *et al.*, Meccanica 15, 9 (1980), doi:10.1007/BF02128236

Kaplan–Yorke Dimension

The exponents also count the directions in which the attractor extends.

Definition (Kaplan–Yorke dimension)

$$D_{\mathrm{KY}} = j + \frac{\lambda_1 + \cdots + \lambda_j}{|\lambda_{j+1}|}$$

j: largest index for which $\lambda_1 + \cdots + \lambda_j \geq 0$

- j-dimensional volumes do not shrink and $(j+1)$-dimensional ones do, so the dimension lies in between
- a limit cycle has $D_{\mathrm{KY}} = 1$, Lorenz-63 2.06, a fractal a little more than a surface
- D_{KY} estimates how many variables a forecasting model must resolve

J. L. Kaplan, J. A. Yorke, Lect. Notes Math. 730, 204 (1979), doi:10.1007/BFb0064319

Stochastic Processes

In a stochastic signal each new value contains a random part that no model can know in advance.

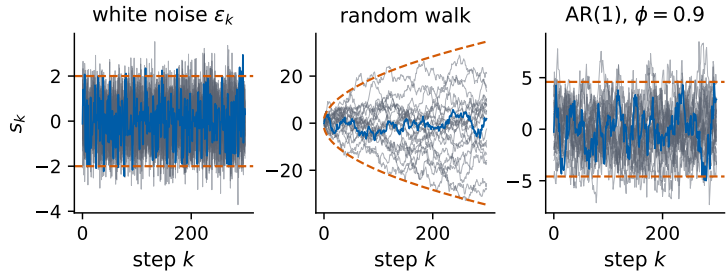
Definition (Three basic processes)

$$\begin{aligned} \text{white noise} \quad & s_k = \varepsilon_k \\ \text{random walk} \quad & s_k = s_{k-1} + \varepsilon_k \\ \text{AR(1)} \quad & s_k = \phi\, s_{k-1} + \varepsilon_k, \quad |\phi| < 1 \end{aligned}$$

ε_k: independent random numbers with mean 0 and variance σ^2; ϕ: memory coefficient

- **stationary**: mean and correlations $\mathrm{cov}(s_k, s_{k+n})$ do not depend on k
- random walk: discrete Brownian motion, variance $k\sigma^2$, not stationary
- AR(1): discrete Ornstein–Uhlenbeck process (velocity with friction), variance $\sigma^2/(1-\phi^2)$ G. E. Uhlenbeck, L. S. Ornstein, Phys. Rev. 36, 823 (1930), doi:10.1103/PhysRev.36.823

Realisations of the Three Processes



20 realisations (grey, one blue), $\sigma = 1$, $\phi = 0.9$; dashed ± 2 standard deviations of the process.

- best forecasts: the mean for white noise, the last value for the random walk, $\phi^\tau s_k$ for AR(1)
- the error of the best forecast stays because of the future ε, for chaos it comes from the initial error

4

Tools for Characterising a Signal



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Autocorrelation Function

The first question about a signal is how long it remembers its past.

Definition (Autocorrelation function (ACF))

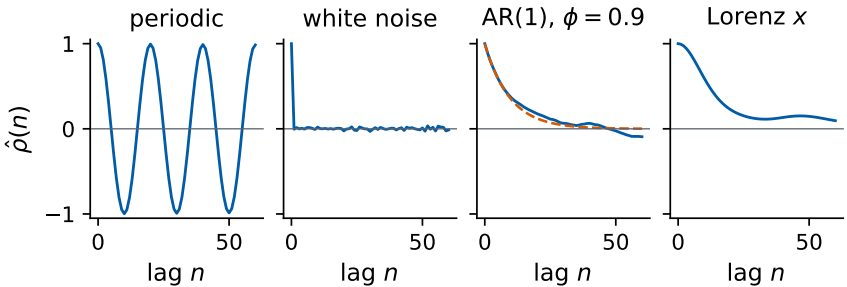
$$\hat{\rho}(n) = \frac{\sum_{k=1}^{N-n} (s_k - \bar{s})(s_{k+n} - \bar{s})}{\sum_{k=1}^{N} (s_k - \bar{s})^2}$$

n: lag in samples; $\bar{s}$: mean of the record; $\hat{\rho}(0) = 1$, $|\hat{\rho}(n)| \leq 1$

- $\hat{\rho}(n)$ estimates the correlation of a stationary process, $\mathrm{cov}(s_k, s_{k+n})/\mathrm{var}(s_k)$
- a correlation time is the first lag at which $\hat{\rho}$ drops below $1/e$, or the first zero
- the ACF only sees linear dependence between s_k and s_{k+n}

G. E. P. Box, G. M. Jenkins, G. C. Reinsel, G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed., Wiley (2015)

ACF of Four Kinds of Signals



$\hat{\rho}(n)$ from 4000 samples; dashed the exact $\rho(n) = \phi^n$ of AR(1).

- periodic: the correlation returns after every period and never decays
- white noise forgets at once, AR(1) exponentially, Lorenz x more slowly and without returning

Power Spectral Density

The power spectral density (PSD) shows at which frequencies the variance of the signal sits.

Definition (Periodogram)

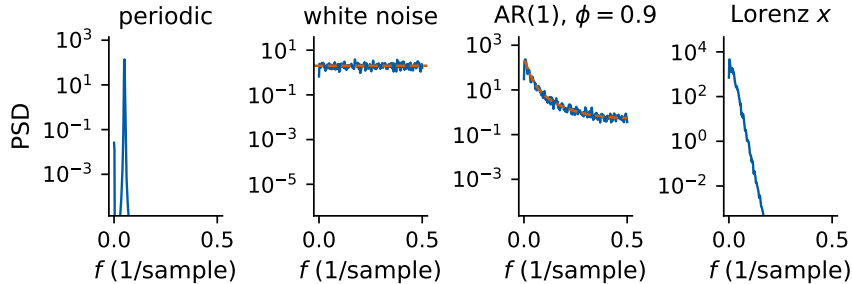
$$X_q = \sum_{k=0}^{N-1} (s_k - \bar{s})\, e^{-2\pi i\, qk/N}, \qquad \hat{S}(f_q) = \frac{2\Delta t}{N} |X_q|^2, \qquad f_q = \frac{q}{N\Delta t}$$

X_q: discrete Fourier transform (FFT); one-sided, $0 < f_q < 1/(2\Delta t)$

- PSD and ACF are a Fourier pair (Wiener–Khinchin), and the area under the PSD is the variance N. Wiener, Acta Math. 55, 117 (1930); A. Khintchine, Math. Ann. 109, 604 (1934)
- in practice we average periodograms of overlapping segments (Welch method, Lab 033)

P. D. Welch, IEEE Trans. Audio Electroacoust. 15, 70 (1967), doi:10.1109/TAU.1967.1161901

Spectra of Four Kinds of Signals

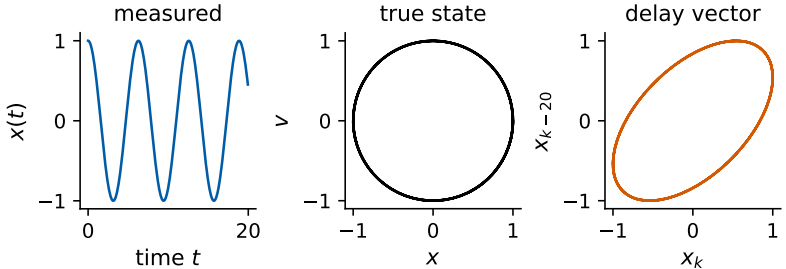


Welch PSD on a logarithmic axis; dashed the exact spectra, $2\sigma^2$ for white noise, $2\sigma^2/|1-\phi e^{-2\pi i f}|^2$ for AR(1).

- a periodic signal puts its power into one line at $f = 1/P$
- white noise is flat, AR(1) and Lorenz x are broadband, so ACF and PSD alone do not separate chaos from noise

Phase Space from One Coordinate

A harmonic oscillator has the state (x, v), but we measure only the position $x(t)$.



- the velocity is hidden in the past, $v_k \approx (x_k - x_{k-1})/\Delta t$
- so the pair $(x_k, x_{k-\ell})$ traces a deformed copy of the true orbit in (x, v)

Lorenz from x Alone

The Lorenz equations can be solved for the unmeasured coordinates.

$$\dot{x} = \sigma(y - x) \;\Rightarrow\; y = x + \frac{\dot{x}}{\sigma}, \qquad \dot{y} = x(\rho - z) - y \;\Rightarrow\; z = \rho - \frac{\dot{y} + y}{x}$$

- $(x, \dot{x}, \ddot{x})$ fix the full state (x, y, z), because $\dot{y} = \dot{x} + \ddot{x}/\sigma$
- derivatives are differences of past values, so a window $(x_k, x_{k-\ell}, x_{k-2\ell})$ carries the state
- delays work better than derivatives on noisy data

N. H. Packard, J. P. Crutchfield, J. D. Farmer, R. S. Shaw, Phys. Rev. Lett. 45, 712 (1980), doi:10.1103/PhysRevLett.45.712

Takens' Theorem

Definition (Delay vector, delay ℓ, dimension m)

$$\mathbf{y}_k = \left(s_k,\ s_{k-\ell},\ \ldots,\ s_{k-(m-1)\ell}\right) \in \mathbb{R}^m$$

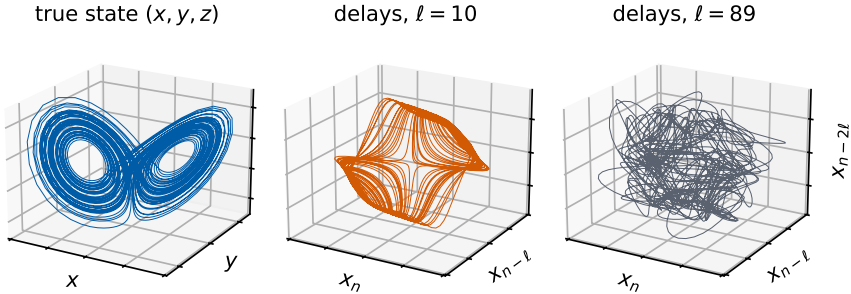
Takens' theorem, in words

One generic measured quantity and $m > 2d$ delays determine the state of a d-dimensional system one-to-one ($m > 2d_A$ for an attractor of dimension d_A).

- why $2d$: the shadow of a curve on a wall can cross itself, a curve in space generically does not
- so a forecast is a function of the window, $s_{k+1} = f(\mathbf{y}_k)$

F. Takens, Lect. Notes Math. 898, 366 (1981), doi:10.1007/BFb0091924; T. Sauer, J. A. Yorke, M. Casdagli, J. Stat. Phys. 65, 579 (1991), doi:10.1007/BF01053745

The Lorenz Attractor Rebuilt from $x(t)$ Alone



True state (x, y, z) and delay vectors $(x_k, x_{k-\ell}, x_{k-2\ell})$ with $\ell = 10$ and $\ell = 89$.

- with $\ell = 10$ (chosen in Section 5) the two lobes reappear as a smoothly deformed image
- with $\ell = 89$, the zero of the ACF, the coordinates are almost unrelated and the shape is lost

5

Signal Characteristics before Forecasting



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Lorenz-63: A Toy Model of the Weather

Edward Lorenz, a meteorologist at MIT, asked how far ahead the weather can be predicted.

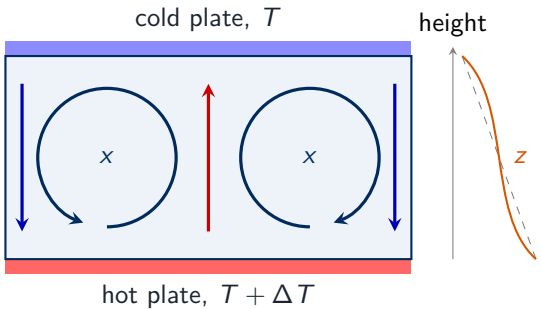
- he took Saltzman's equations of a fluid layer heated from below (Rayleigh–Bénard convection, the process that drives atmospheric circulation) and kept only three Fourier modes
- the heating ρ sets the regime: for $\rho < 1$ heat is conducted and the fluid rests, for $1 < \rho < 24.74$ steady rolls turn, at $\rho = 28$ the rolls flip direction irregularly
- in an earlier, larger weather model he restarted a run from rounded numbers and saw the forecast drift away, the effect of the frame *Two Nearby States Separate*

E. N. Lorenz, J. Atmos. Sci. 20, 130 (1963), doi:10.1175/1520-0469(1963)020<0130:DNF>2.0.CO;2;

B. Saltzman, J. Atmos. Sci. 19, 329 (1962), doi:10.1175/1520-0469(1962)019<0329:FAFCAA>2.0.CO;2;

E. N. Lorenz, *The Essence of Chaos*, University of Washington Press (1993)

Lorenz-63: The Physical Picture



- x: speed of the convection rolls; its sign is the sense of rotation
- y: temperature difference between the warm rising (red) and the cold sinking (blue) fluid
- z: deviation of the vertical temperature profile (orange) from the linear profile of pure conduction (dashed)

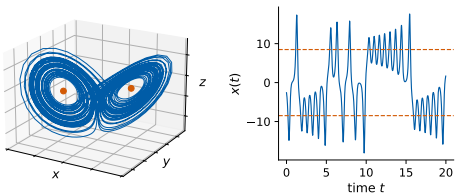
Lorenz-63: Variables and Parameters

	meaning	value
x	intensity of the convective circulation, the speed of the rolls	
y	temperature difference between rising and sinking fluid	
z	distortion of the vertical temperature profile from the linear one	
σ	Prandtl number, viscosity over thermal diffusivity	10
ρ	heating, the Rayleigh number in units of its critical value	28
β	geometry, set by the aspect ratio of the convection rolls	$8/3$

Changing ρ changes the physics, from rest to steady rolls to chaos. We forecast the signal x, the speed of the rolls.

Lorenz-63: Equations and Attractor

$$\dot{x} = \sigma(y-x), \quad \dot{y} = x(\rho-z)-y, \quad \dot{z} = xy-\beta z$$



- left the attractor in (x, y, z), right the signal $x(t)$
- the rolls turn one way, speed up, and flip irregularly to the other way
- we observe only $s_k = x(k\Delta t)$, $\Delta t = 0.02$

E. N. Lorenz, J. Atmos. Sci. 20, 130 (1963), doi:10.1175/1520-0469(1963)020<0130:DNF>2.0.CO;2

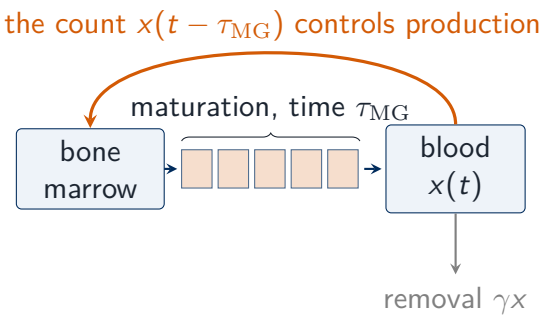
Mackey–Glass: Regulation of Blood Cells

Mackey and Glass studied physiological control systems with a delay.

- the bone marrow produces blood cells in response to the number of circulating cells
- a new cell needs the time τ_{MG} to mature, so production responds to the count τ_{MG} ago
- a longer delay turns the steady state into oscillations and then chaos; Mackey and Glass called such abnormal rhythms dynamical diseases
- the attractor dimension grows with the delay (Farmer), as our D_{KY} does

M. C. Mackey, L. Glass, Science 197, 287 (1977), doi:10.1126/science.267326; J. D. Farmer, Physica D 4, 366 (1982), doi:10.1016/0167-2789(82)90042-2; L. Glass, M. C. Mackey, *From Clocks to Chaos: The Rhythms of Life*, Princeton University Press (1988)

Mackey–Glass: A Feedback Loop with a Delay



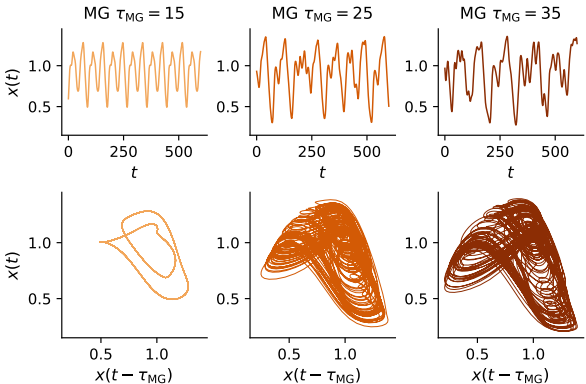
- cells leave the marrow only after maturing for the time τ_{MG}, and are removed from the blood at the rate γ
- the marrow reacts to the count of τ_{MG} ago, so it keeps producing after the count is already high, and overshoots
- a short delay is corrected in time, a long one gives oscillations and chaos

Mackey–Glass: Variables and Parameters

	meaning	value
$x(t)$	concentration of circulating blood cells	
β	production rate, driven by the count τ_{MG} ago	0.2
n	steepness of the feedback	10
γ	removal rate of cells	0.1
τ_{MG}	maturation delay	15, 25, 35

With $n = 10$ production first grows and then falls with the past count, a negative feedback. The delay τ_{MG} is the knob we turn: periodic at 15, weak chaos at 25, developed chaos at 35.

Mackey–Glass: a Delayed Feedback



Blood-cell model, x: cell count

$$\dot{x} = \underbrace{\beta \frac{x(t-\tau_{\mathrm{MG}})}{1 + x(t-\tau_{\mathrm{MG}})^{10}}}_{\text{production}} - \underbrace{\gamma x}_{\text{death}}$$

$\beta = 0.2$, $\gamma = 0.1$; upper panels $x(t)$, lower panels $x(t - \tau_{\mathrm{MG}})$ against $x(t)$

- the late reaction overshoots: periodic at $\tau_{\mathrm{MG}} = 15$, chaotic at 25 and 35

M. C. Mackey, L. Glass, Science 197, 287 (1977), doi:10.1126/science.267326

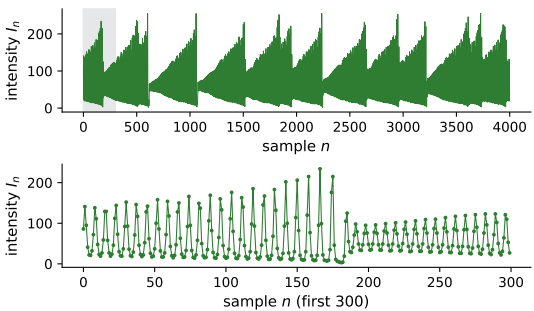
Santa Fe: A Laser Described by the Lorenz Equations

The record is the intensity of a far-infrared laser filled with ammonia gas (NH_3), measured by Hübner, Abraham and Weiss.

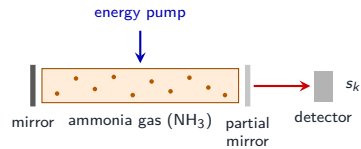
- a single-mode laser is described by three variables, the field E, the polarisation P of the gas and the population inversion D
- Haken showed that these laser equations become the Lorenz equations after rescaling, with the pump strength in the role of the heating ρ
- so the measured laser and the convection model share one mathematical structure
- the record became data set A of the Santa Fe time-series prediction competition, hence its name

H. Haken, Phys. Lett. A 53, 77 (1975), doi:10.1016/0375-9601(75)90353-9; U. Hübner, N. B. Abraham, C. O. Weiss, Phys. Rev. A 40, 6354 (1989), doi:10.1103/PhysRevA.40.6354; A. S. Weigend, N. A. Gershenfeld (eds.), *Time Series Prediction: Forecasting the Future and Understanding the Past* (1994)

Santa Fe: a Chaotic Laser



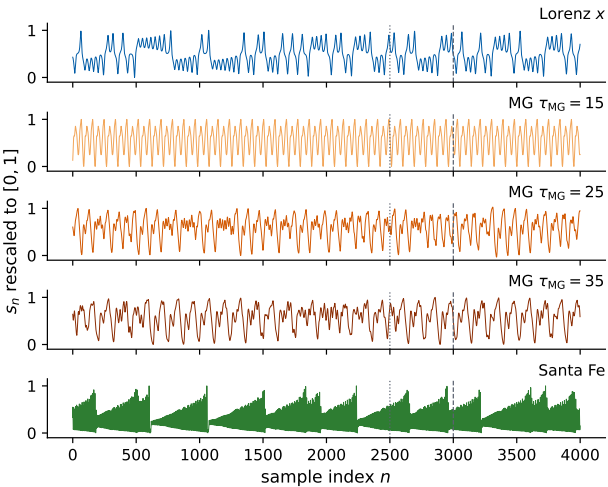
Brightness of an ammonia gas laser



- energy pumped into the gas is stored, then released in a flash
- flashes grow, then collapse, never repeating exactly
- a measured record, 8-bit integers, no equations used

U. Hübner, N. B. Abraham, C. O. Weiss, Phys. Rev. A 40, 6354 (1989), doi:10.1103/PhysRevA.40.6354

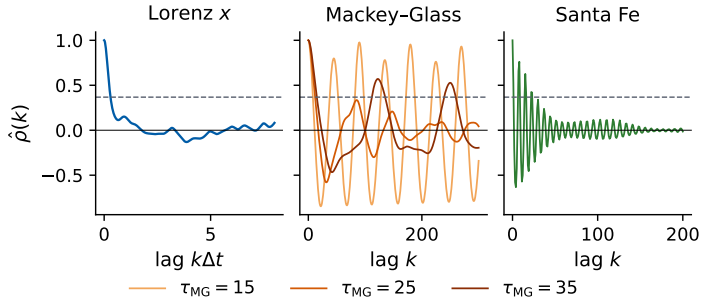
The Signals and the Data Split



Lorenz x, Mackey–Glass (MG) at three delays and Santa Fe; each s_k rescaled to $[0, 1]$ with the minimum and maximum of its first 3000 samples.

- $N = 4000$ samples per signal
- training block up to sample 2500 (dotted), validation up to 3000 (dashed), then the test block
- the models of this lecture are trained and scored on these five records

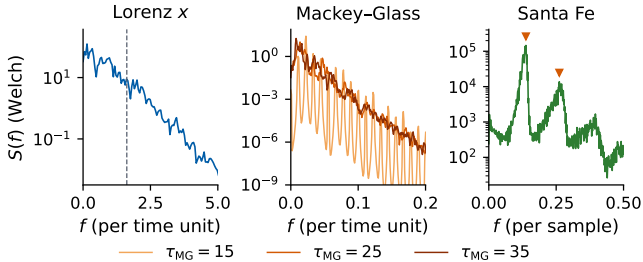
Memory: Autocorrelation of the Signals



Sample ACF $\hat{\rho}(n)$, dashed line at $1/e$; Lorenz x against $n\Delta t$.

- Lorenz x falls below $1/e$ after 15 samples and crosses zero after 89
- the periodic Mackey–Glass ACF ($\tau_{\mathrm{MG}} = 15$) never decays, the chaotic ones decay slowly, Santa Fe oscillates with the flashes

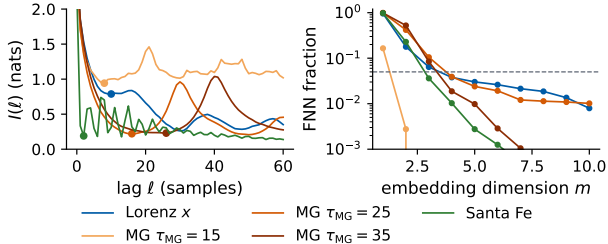
Frequencies: Spectra of the Signals



Welch PSD, segments of 1024 samples; triangles mark peaks a decade above their surroundings.

- MG 15 has lines at multiples of one frequency, MG 25 and 35 broad peaks on a broadband background
- Lorenz x is broadband without peaks, Santa Fe has a sharp peak at the flash period of 7 to 8 samples

State: Delay and Dimension of the Signals



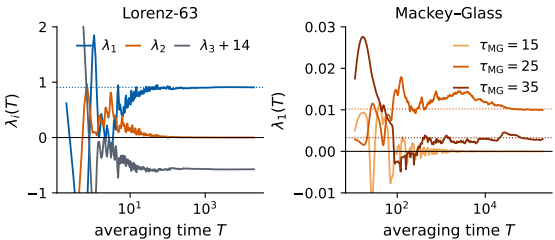
A. M. Fraser, H. L. Swinney, Phys. Rev. A 33, 1134 (1986)

M. B. Kennel, R. Brown, H. D. I. Abarbanel, Phys. Rev. A 45, 3403 (1992)

Left the mutual information of s_k and $s_{k+\ell}$, its first minimum gives ℓ; right the fraction of false neighbours against m, 5% dashed.

- (ℓ, m) is (10, 4) for Lorenz, (8, 2), (16, 4), (26, 4) for MG 15, 25, 35, (2, 3) for Santa Fe
- a closed curve (MG 15) needs $m = 2$, the chaotic signals 3 to 4

Chaos: Lyapunov Exponents and Dimension



Exponents from the equations, running averages over the integration time; MG is sampled once per time unit.

M. T. Rosenstein *et al.*, Physica D 65, 117 (1993)

signal	λ_1 per time unit	D_{KY}
Lorenz x	0.907	2.06
MG 15	≈ 0	1.00
MG 25	0.0099	3.16
MG 35	0.0029	3.41

- MG 15 is a limit cycle
- chaotic MG: more dimensions than Lorenz, slower error growth per sample (Lorenz $\lambda_1 \Delta t = 0.018$)
- Santa Fe from the record alone, $\lambda_1 \approx 0.029$ per sample

Consequences for Forecasting

- The tested **horizons** $\tau \leq 50$ are about one Lyapunov time for Lorenz x ($T_\lambda = 55$ samples), well within one for Mackey–Glass ($T_\lambda = 101$ and 340 samples), and more than one for the laser ($T_\lambda \approx 34$ samples).
- A delay vector covers a **window** of $(m-1)\ell + 1$ samples, 31 for Lorenz x, 49 and 79 for $\tau_{\mathrm{MG}} = 25, 35$, 5 for Santa Fe, so a model needs at least this many past values.
- The **limit cycle** MG 15 has no finite T_λ, so all horizons can be forecast.
- $D_{\mathrm{KY}} > 2$: a curved, folded attractor, which a linear model follows only over short horizons.

6

Linear Models



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Forecast as a Linear Combination of Features

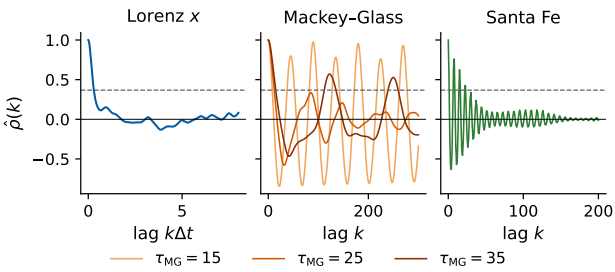
All models of Lectures 7 and 8 predict the future value as a linear combination of features computed from the past,

$$\hat{s}_{k+\tau} = \sum_{i=1}^{M} w_i \, \varphi_i(s_k, s_{k-1}, \dots) = \mathbf{w}^\top \boldsymbol{\varphi}_k$$

- the features φ_i are in general nonlinear functions of the past values
- the weights w_i are trained, and the models differ only in the features

model	features $\boldsymbol{\varphi}_k$
AR	the past values $s_k, \dots, s_{k-p+1}$ (linear)
MA, ARMA	past values and past prediction errors
NVAR	past values and their products
recurrent networks (Lecture 8)	the state of a network driven by the signal

The Past Predicts the Future



The autocorrelation function of our signals stays large over many lags.

- a large $\rho(n)$ means that s_{k-n} carries information about s_k
- so the simplest forecast weighs the last few values and adds them up
- this is autoregression, the first of four classical linear models

Autoregression: Weighted Sum of the Last p Values

The forecast of the next value is a weighted sum of the last p measured values,

$$\hat{s}_{k+1} = c + \sum_{i=1}^{p} \phi_i \, s_{k+1-i} = c + \phi_1 s_k + \phi_2 s_{k-1} + \cdots + \phi_p s_{k-p+1}$$

- **trainable parameters**: the constant c and the weights $\phi_1, \ldots, \phi_p$
- the order p is chosen by us, on the validation block or by the Akaike information criterion
- what the forecast misses, $s_{k+1} - \hat{s}_{k+1}$, is treated as white noise; with $p = 1$ this is the AR(1) process of the stochastic part

G. U. Yule, Phil. Trans. R. Soc. A 226, 267 (1927), doi:10.1098/rsta.1927.0007; H. Akaike, IEEE Trans. Autom. Control 19, 716 (1974), doi:10.1109/TAC.1974.1100705

Fitting an AR Model

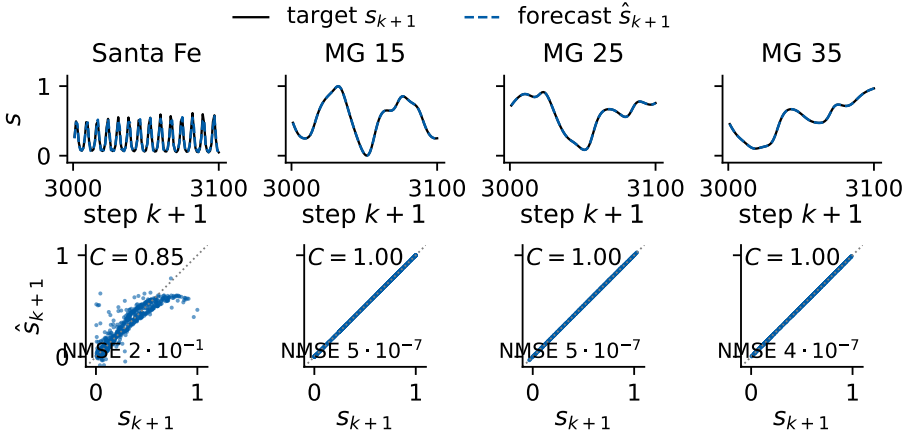
The parameters enter linearly, so fitting is one least-squares problem. Each row of X is one window of past values, and $\mathbf{y}$ holds the value that followed it.

$$X = \begin{pmatrix} 1 & s_{p-1} & \cdots & s_0 \\ \vdots & \vdots & & \vdots \\ 1 & s_{n-2} & \cdots & s_{n-p-1} \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} s_p \\ \vdots \\ s_{n-1} \end{pmatrix}, \quad \begin{pmatrix} \hat{c} \\ \hat{\phi}_1 \\ \vdots \\ \hat{\phi}_p \end{pmatrix} = \arg\min_{\beta} \|\mathbf{y} - X\beta\|^2$$

- no gradient descent is needed, the solution follows from $X^\top X\beta = X^\top \mathbf{y}$
- our JAX implementation agrees with `statsmodels` to round-off

S. Seabold, J. Perktold, Proc. 9th Python in Science Conf., 92 (2010), doi:10.25080/Majora-92bf1922-011

One-Step AR Forecasts



Mackey–Glass is followed almost perfectly. On Santa Fe the points bend away from the diagonal at the high flashes, so something is missing.

Learning from Your Own Mistakes

AR uses only the measured values. A forecaster can also look at how wrong it was.

- yesterday the forecast was too low, so today we raise it a little
- as an analogy, a thermostat corrects the heating from the recent temperature errors
- the prediction errors $e_k = s_k - \hat{s}_k$ become the input of the forecast

A forecast built from the recent errors is a moving-average (MA) model.

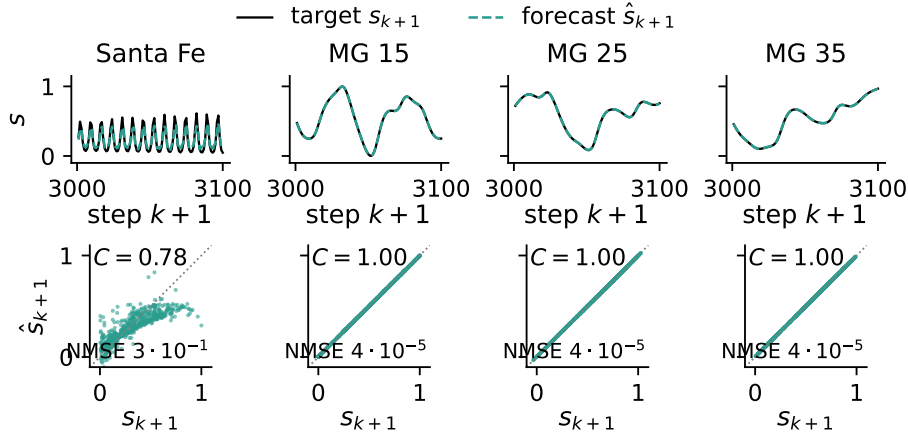
Moving Average: Weighted Sum of the Last q Prediction Errors

The forecast corrects the mean by the errors the model made in the last q steps,

$$e_k = s_k - \hat{s}_k, \qquad \hat{s}_{k+1} = \mu + \sum_{j=1}^{q} \theta_j \, e_{k+1-j}$$

- **trainable parameters**: the mean μ and the weights $\theta_1, \ldots, \theta_q$
- the errors are computed step by step along the record; they depend on the parameters, so the fit maximises the likelihood
- an error older than q steps no longer enters, so beyond q steps ahead the forecast is the mean μ

One-Step MA Forecasts



MA(8) for each signal. Mackey–Glass is still followed closely, while on Santa Fe MA is the weakest model.

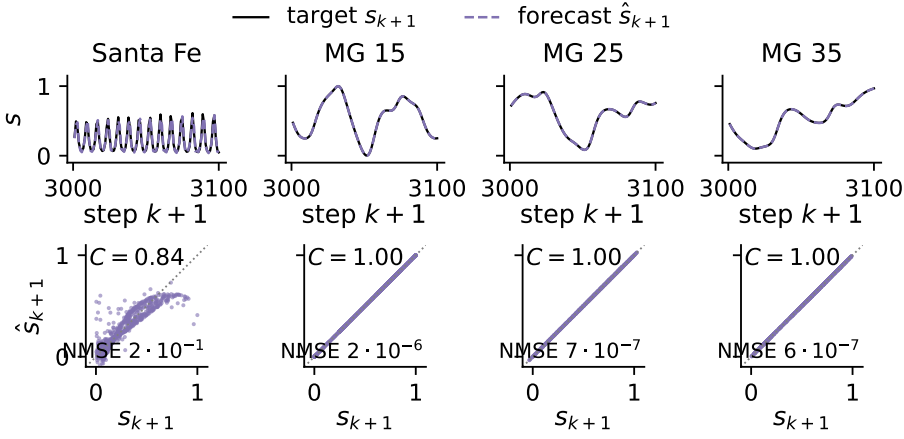
ARMA: Both Sums

ARMA combines the past values and the past prediction errors,

$$\hat{s}_{k+1} = c + \sum_{i=1}^{p} \phi_i \, s_{k+1-i} + \sum_{j=1}^{q} \theta_j \, e_{k+1-j}$$

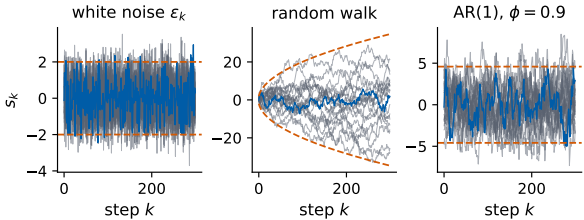
- **trainable parameters**: c, the AR weights ϕ_i and the MA weights θ_j
- the orders p and q are chosen by us
- a few MA terms can replace many AR terms, so ARMA often needs fewer parameters

One-Step ARMA Forecasts



Orders chosen by the Akaike information criterion. On these deterministic signals the MA terms add nothing over AR. H. Akaike, IEEE Trans. Autom. Control 19, 716 (1974)

A Signal Without a Fixed Mean



The random walk of the stochastic part (middle panel) wanders away and has no fixed mean.

- AR, MA and ARMA assume a signal that fluctuates around a fixed mean
- the steps of the random walk, $s_k - s_{k-1} = \varepsilon_k$, do fluctuate around zero
- so we forecast the steps and add them up

ARIMA: Forecasting the Differences

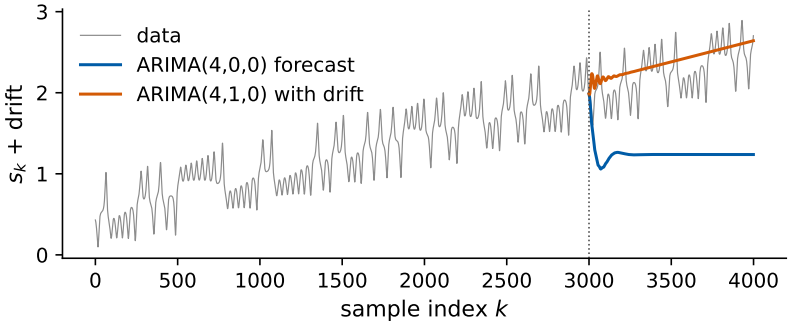
ARIMA applies ARMA to the differences of the signal.

$$\Delta s_k := s_k - s_{k-1}, \qquad \Delta^d s_k := \Delta(\Delta^{d-1} s_k), \qquad z_k := \Delta^d s_k$$

1. difference the signal d times
2. forecast the differences with ARMA(p, q),
   $\hat{z}_{k+1} = c + \sum_{i=1}^{p} \phi_i z_{k+1-i} + \sum_{j=1}^{q} \theta_j e_{k+1-j}$
3. add back, $\hat{s}_{k+1} = s_k + \hat{z}_{k+1}$ for $d = 1$, $\hat{s}_{k+1} = 2s_k - s_{k-1} + \hat{z}_{k+1}$ for $d = 2$

The trainable parameters are those of ARMA, and d is chosen by us. G. E. P. Box, G. M. Jenkins, G. C. Reinsel, G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed., Wiley (2015)

Differencing Removes a Drift



Lorenz x plus a linear drift; each model forecasts the whole test block from its first step.

- without differencing ($d = 0$) the forecast relaxes to the mean
- with $d = 1$ it continues the trend, but neither follows the oscillation

Compact Notation: The Backshift Operator

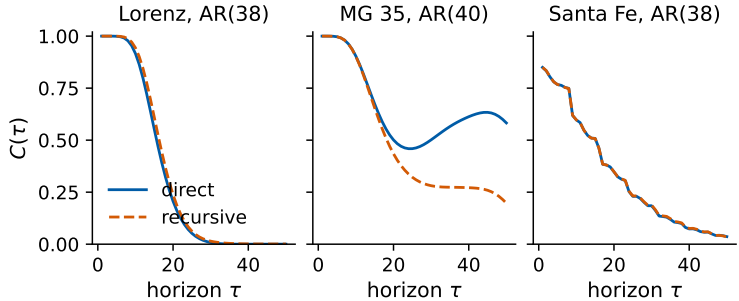
A compact notation for shifts in time makes long models readable.

$$Bs_k := s_{k-1}, \qquad B^2 s_k = s_{k-2}, \qquad (1-B)\, s_k = s_k - s_{k-1}$$

- AR(2), $s_k = \phi_1 s_{k-1} + \phi_2 s_{k-2} + \varepsilon_k$, becomes $(1 - \phi_1 B - \phi_2 B^2)\, s_k = \varepsilon_k$
- in general we collect the weights into two polynomials in B,

$$\Phi_p(B) := 1 - \phi_1 B - \cdots - \phi_p B^p, \qquad \Theta_q(B) := 1 + \theta_1 B + \cdots + \theta_q B^q$$

Direct and Recursive Linear Forecasts



$C(\tau)$ on the test block; recursive iterates one AR model, direct fits one AR map per τ.

- an iterated linear model relaxes to its mean
- direct wins on Mackey–Glass, on Lorenz and Santa Fe both lose the signal within about 20 steps

Four Linear Models in One View

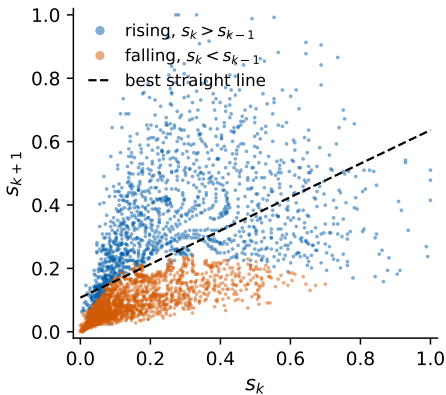
Each model predicts the next value $\hat{s}_{k+1}$ from the past.

- **AR(p)**, autoregression: a weighted sum of the last p values
- **MA(q)**, moving average: a weighted sum of the last q prediction errors $e_k = s_k - \hat{s}_k$
- **ARMA(p, q)**: both sums
- **ARIMA(p, d, q)**, integrated ARMA, for signals with a trend: forecast the differences of the signal with ARMA and add them back

All four are linear in the past values, and their coefficients are fitted to the training data.

G. E. P. Box, G. M. Jenkins, G. C. Reinsel, G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed., Wiley (2015); P. J. Brockwell, R. A. Davis, *Introduction to Time Series and Forecasting*, Springer (2016), doi:10.1007/978-3-319-29854-2

What a Linear Model Cannot Follow



The Santa Fe laser, next value s_{k+1} against the present value s_k on the training block.

- the same s_k is followed by different s_{k+1} on a rising and on a falling flash; one more lag tells the two apart
- but each branch is curved, and a linear model is a straight line (with more lags, a flat plane) through the cloud
- to follow the fold the forecast must be a nonlinear function of the past

7

Nonlinear Vector Autoregression



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Nonlinear Features of the Window

To follow the fold we keep the window of past values and add their products as further features.

Definition (Nonlinear vector autoregression (NVAR))

$$\varphi_k = \big(1,\ \underbrace{s_k, s_{k-1}, \ldots, s_{k-L+1}}_{\text{linear}},\ \underbrace{s_{k-i}\, s_{k-j}\ (0 \le i \le j \le L-1)}_{\text{quadratic}}\big), \qquad \hat{s}_{k+\tau} = \mathbf{w}_\tau^{\top} \varphi_k$$

- **trainable parameters**: the readout weights $\mathbf{w}_\tau$, one vector for every lead τ
- the features are fixed, so training is again one least-squares problem
- L and a stride ℓ (every ℓ-th value) are chosen by us

D. J. Gauthier, E. Bollt, A. Griffith, W. A. S. Barbosa, Nat. Commun. 12, 5564 (2021),
doi:10.1038/s41467-021-25801-2

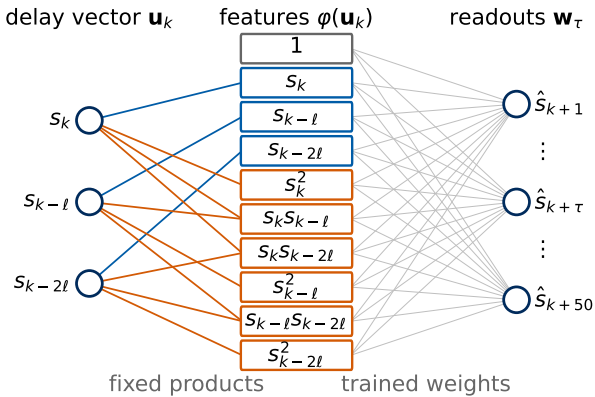
NVAR with Two Past Values

For $L = 2$ the forecast is written out in full,

$$\hat{s}_{k+1} = w_0 + \underbrace{w_1 s_k + w_2 s_{k-1}}_{\text{as in AR(2)}} + \underbrace{w_3 s_k^2 + w_4 s_k s_{k-1} + w_5 s_{k-1}^2}_{\text{new: products}}$$

- with the products the forecast can curve: the effect of s_k depends on the values themselves, so it can follow a folded cloud instead of a flat plane
- a window of L values gives $1 + L + L(L+1)/2$ features, 6 for $L = 2$ and 153 for $L = 16$
- the number of features grows quickly with L, so the weights need a penalty (next frames)

NVAR Feature Map



Feature map for $L = 3$: each input feeds its linear feature (blue) and the products that contain it (orange).

- only the weights to the readouts are trained
- the number of features grows as L^2, 153 at $L = 16$

Ridge Readout

With hundreds of features, plain least squares fits the noise.

Definition (Ridge regression)

$$\mathbf{w}_\tau = \arg\min_{\mathbf{w}} \sum_k \left(s_{k+\tau} - \mathbf{w}^\top \varphi_k\right)^2 + \lambda \|\mathbf{w}\|^2 = \left(X^\top X + \lambda I\right)^{-1} X^\top \mathbf{y}$$

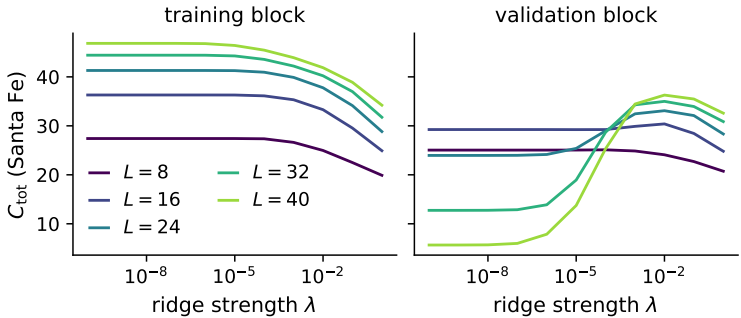
$\lambda > 0$: ridge strength; X: features φ_k as rows; $\mathbf{y}$: targets

- the penalty keeps the weights small when features are nearly redundant
- ReservoirPy builds the same model as `NVAR(delay, order, strides) » Ridge(ridge)`

A. E. Hoerl, R. W. Kennard, Technometrics 12, 55 (1970), doi:10.1080/00401706.1970.10488634;

N. Trouvain *et al.*, ICANN 2020, doi:10.1007/978-3-030-61616-8_40, reservoirpy.readthedocs.io

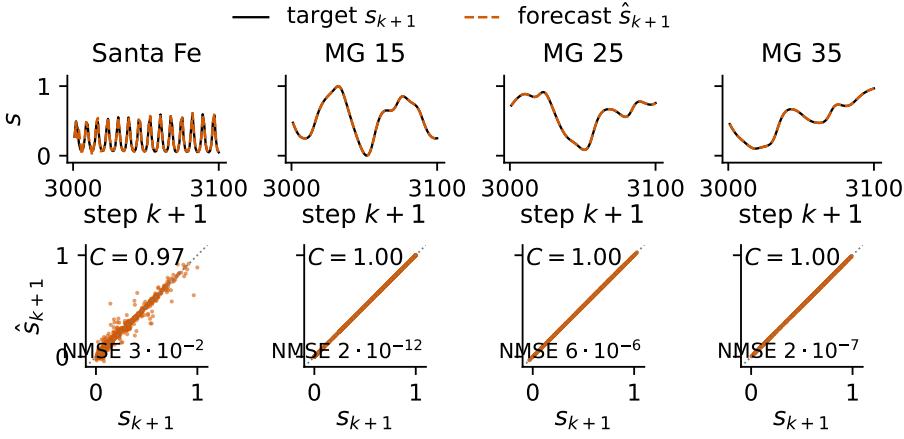
Overfitting and Regularisation



C_{tot} on Santa Fe against λ for windows $L = 8, \ldots, 40$; left training, right validation block.

- on training data longer windows and smaller λ always look better
- on validation data a long window with tiny λ collapses, so L and λ are chosen there

One-Step NVAR Forecasts



L, stride and λ chosen on the validation block. On Santa Fe the points stay near the diagonal also at the high flashes, where the linear models bend away.

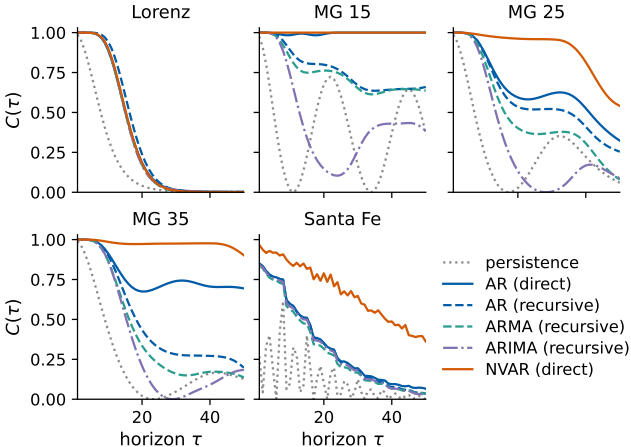
8

Comparison of the Models



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Capacity Curves



$C(\tau)$ on the test block, horizons 1 to 50.

- limit cycle MG 15: direct AR and NVAR near $C = 1$ at all horizons
- chaotic Mackey–Glass: NVAR stays near 1 far beyond the linear models
- Lorenz: every model decays within about 25 samples

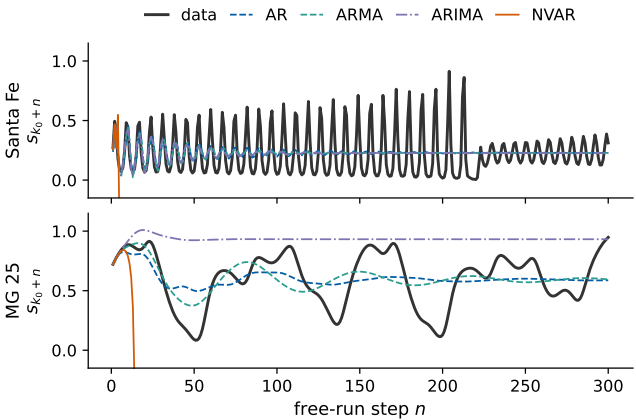
Total Capacity

$C_{\mathrm{tot}} = \sum_{\tau=1}^{50} C(\tau)$ on the test block, at most 50; best per signal in bold.

model	Lorenz	MG 15	MG 25	MG 35	Santa Fe
persistence	8.4	19.5	13.3	11.8	5.5
AR, direct	15.3	49.8	32.6	38.7	16.8
AR, recursive	**16.4**	38.2	29.3	24.6	16.3
ARMA, recursive	15.1	37.1	24.1	20.6	15.3
NVAR, direct	15.3	**50.0**	**44.7**	**48.7**	**33.7**

- every fitted model beats persistence
- NVAR wins on Mackey–Glass and Santa Fe, where the attractor is folded

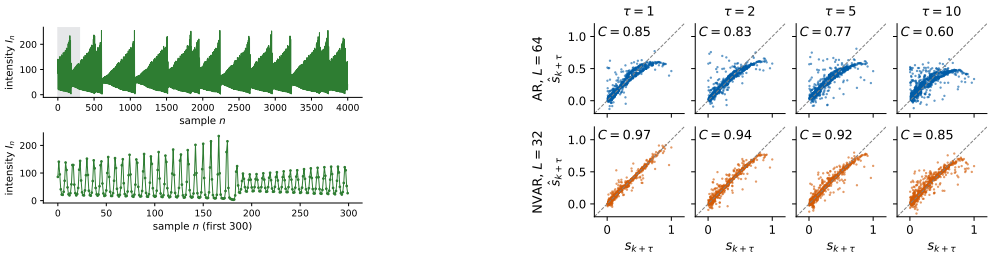
Free Run



Each model runs on its own outputs for 300 steps.

- linear models relax to a constant
- one-step NVAR leaves the data range after 5 steps on Santa Fe
- a good forecast map can be an unstable dynamical model

Limits of Linear Models



Santa Fe: forecast against target, AR (upper row) and NVAR (lower row) at $\tau = 1, 2, 5, 10$.

- AR underestimates the large flashes and its cloud widens with τ, NVAR stays near the diagonal
- NVAR features are chosen by hand, grow fast with L, and remember only a fixed window
- Lecture 8: networks learn the features from data and keep the history in a state

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Summary



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Characterise First

- a task is fixed by its target, a past value (recall) or a future value (forecast), and scored by $C(\tau)$ and NMSE against persistence
- constant, periodic and quasi-periodic signals are trivial to forecast, chaotic and stochastic signals are hard for different reasons
- deterministic chaos has known equations, and the Lyapunov exponent λ_1 sets the prediction horizon $T_{\mathrm{pred}} = \lambda_1^{-1} \ln(\Delta/\delta_0)$
- a stochastic signal contains a new random part at every step, which no model can forecast
- the ACF measures memory, the PSD the frequencies, delay vectors rebuild the state, and D_{KY} counts its directions

Then Forecast

- every model maps past values to features Φ and reads them out linearly
- AR, MA and ARMA are linear models of stochastic processes, and one step ahead they already follow the smooth Mackey–Glass signals
- NVAR adds products of past values and keeps training a ridge regression, and it beats all linear models on the chaotic Mackey–Glass signals and on the laser
- Lecture 8 starts with the Koopman operator, which explains why linear readouts on good features work, and then lets networks produce the features

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References



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